

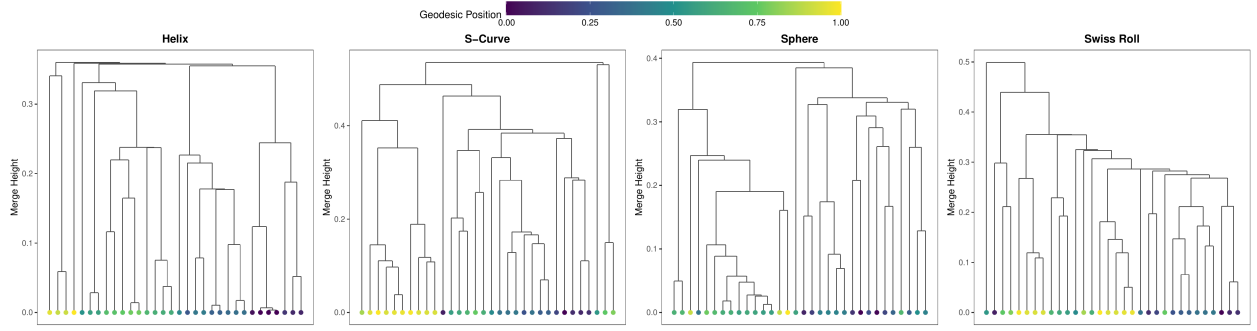


Figure 1: **Single-trial AHC dendrograms for four artificial datasets.** We sample $N = 300$ points from each manifold (Helix, S-Curve, Sphere, Swiss Roll) and build a single AHC tree via random subsampling with subsample size $\psi = 32$ and single linkage. Each leaf represents one of the 32 randomly subsampled data points. Leaf nodes are coloured by their *geodesic position*: arc length for Helix; first intrinsic coordinate for S-Curve and Swiss Roll; azimuthal angle for Sphere, normalised to $[0, 1]$ and mapped to a viridis colour scale.

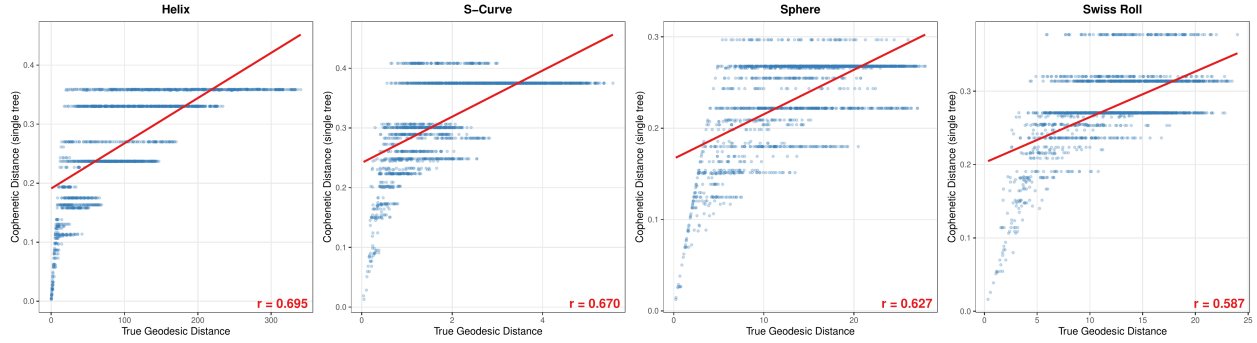


Figure 2: **Cophenetic distance versus true geodesic distance (single AHC tree).** For each manifold ($N = 300$), a single AHC tree is constructed with $\psi = 64$ and single linkage. For every pair of subsampled points ($\binom{64}{2} = 2016$ pairs), we compute the **cophenetic distance (the merge height at which the two points first join in the dendrogram)** and the true geodesic distance. Each dot is one pair; the red line is a linear regression fit, with Pearson r annotated. Pearson r measures *linear* association only. A moderate r does not imply failure to preserve the geodesic *rank ordering*. We report Pearson r as a conservative guarantee.

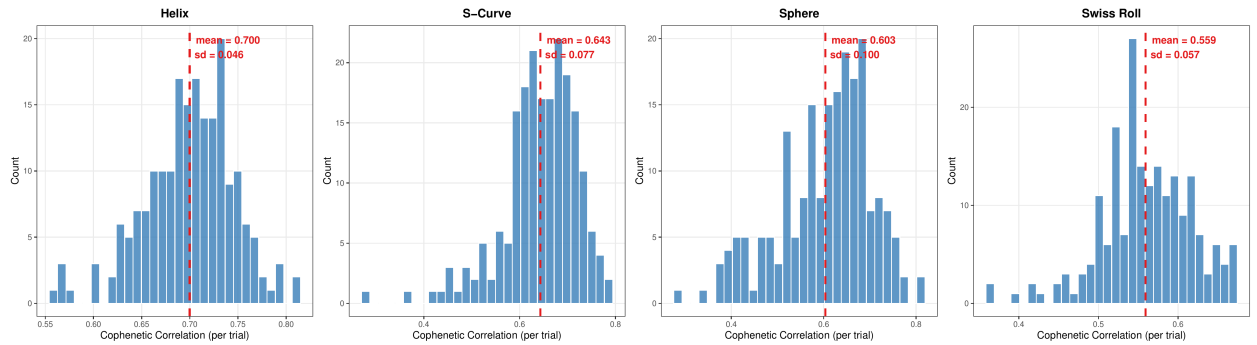


Figure 3: **Distribution of cophenetic correlations across $t = 200$ Monte-Carlo trials.** We repeat the single-tree cophenetic correlation measurement (as in Figure 2) independently $t = 200$ times, each with a random subsample ($\psi = 64$, single linkage). The histogram shows the distribution of Pearson r values across trials; the dashed red line marks the mean. All four manifolds yield consistently positive correlations, the distributions are concentrated well above zero and roughly unimodal.

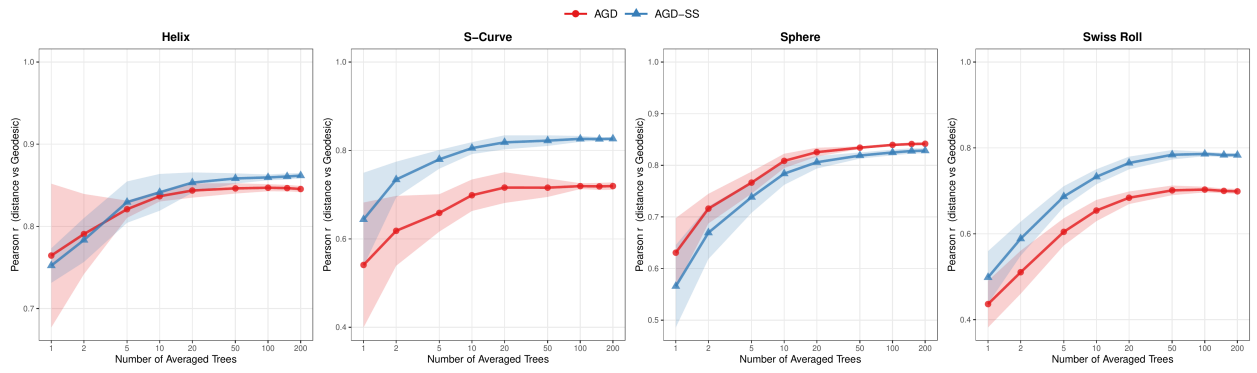


Figure 4: **Aggregation effect: Pearson correlation versus number of averaged trees t .** For each manifold ($N = 300$), we build forests of up to $T = 200$ independent AHC trees ($\psi = 64$, single linkage). Two distance definitions are compared: **AGD** and **AGD-SS**. Solid lines show the mean Pearson r across 10 independent repetitions; shaded bands show standard deviation, quantifying the variance due to random subsampling.

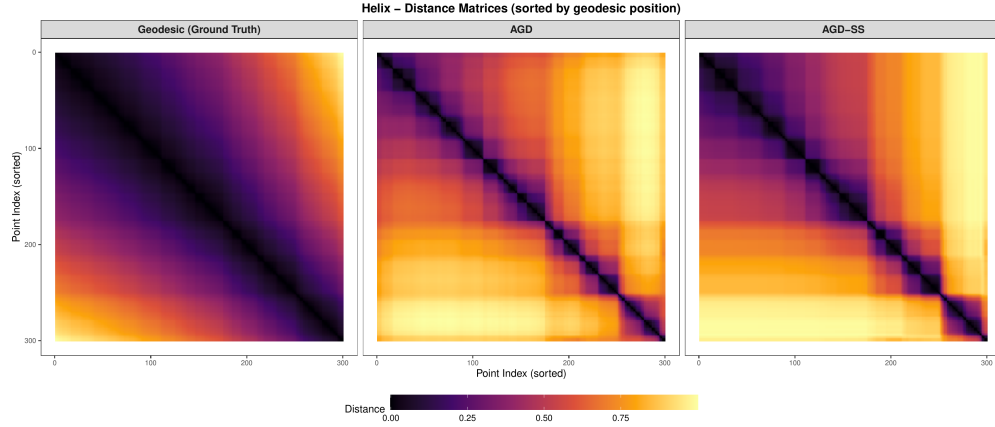


Figure 5: **Distance matrix heatmaps for the Helix manifold.** The full 300×300 pairwise distance matrices are computed using $\psi = 64$ and $t = 200$ trees with single linkage. Rows and columns are sorted by geodesic position (arc length), and each matrix is normalised to $[0, 1]$. Left: ground-truth geodesic distance; Centre: AGD ($r = 0.848$); Right: AGD-SS ($r = 0.863$).

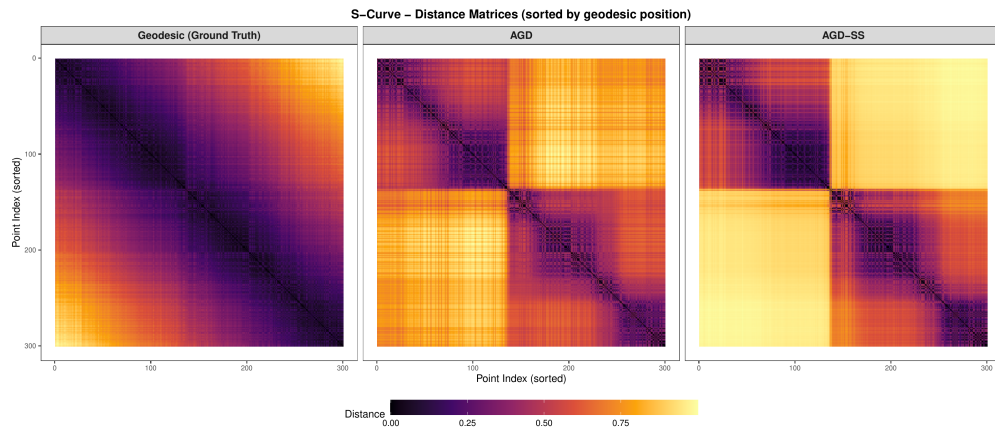


Figure 6: **Distance matrix heatmaps for the S-Curve manifold.** Same protocol as Figure 5: $N = 300$, $\psi = 64$, $t = 200$ trees, single linkage. The geodesic distance is computed via k -NN graph shortest paths ($k = 10$). Matrices are sorted by the first intrinsic coordinate. Left: ground-truth geodesic distance; Centre: AGD ($r = 0.716$); Right: AGD-SS ($r = 0.821$).

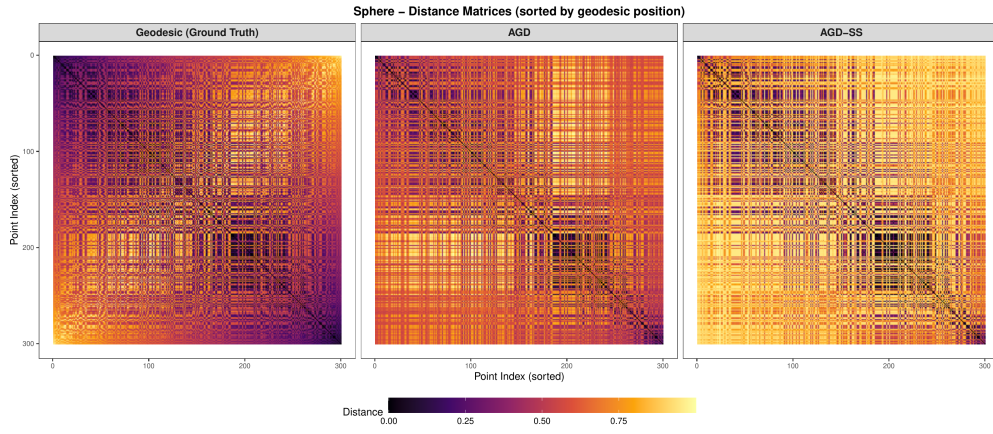


Figure 7: **Distance matrix heatmaps for the Sphere manifold.** Same protocol: $N = 300$, $\psi = 64$, $t = 200$ trees, single linkage. Matrices are sorted by azimuthal angle. Left: ground-truth geodesic distance; Centre: AGD ($r = 0.838$); Right: AGD-SS ($r = 0.827$).

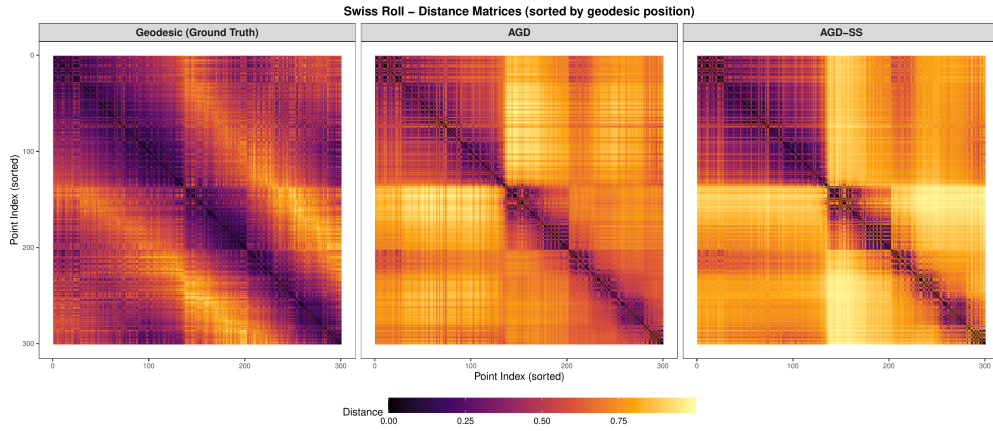


Figure 8: **Distance matrix heatmaps for the Swiss Roll manifold.** Same protocol: $N = 300$, $\psi = 64$, $t = 200$ trees, single linkage. The geodesic distance is computed via k -NN graph shortest paths ($k = 10$); matrices are sorted by the unrolled position. Left: ground-truth geodesic distance; Centre: AGD ($r = 0.676$); Right: AGD-SS ($r = 0.767$). The Swiss Roll is the most challenging manifold due to its tight spiral, where Euclidean-close points from different layers are geodesically distant.

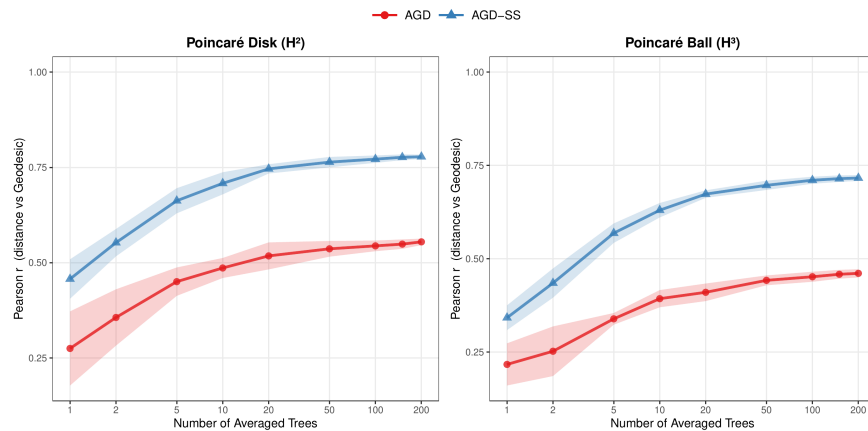


Figure 9: **Aggregation effect on hyperbolic manifolds** ($\psi = 64$, single linkage, 10 repetitions, $\pm\sigma$ bands). Red: AGD; Blue: AGD-SS. On H^2 , AGD-SS outperforms AGD consistently over different number of trees.

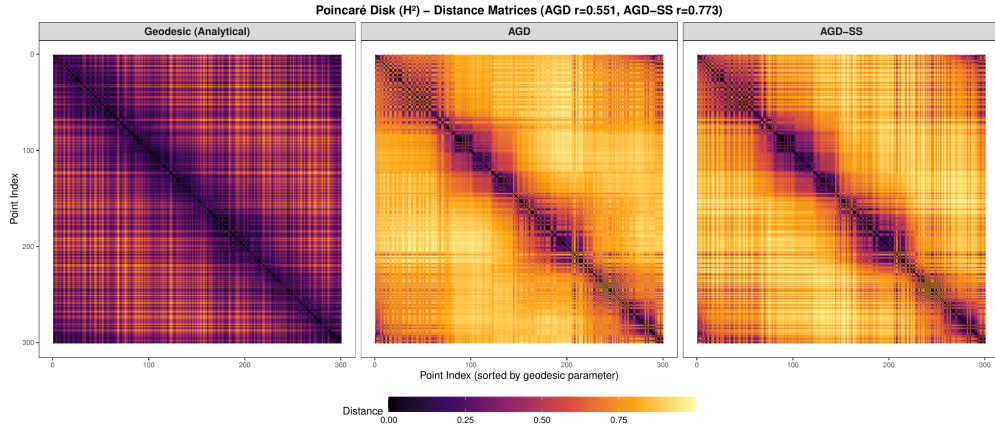


Figure 10: **Distance matrix heatmaps for Poincaré Disk (H2)**. 300×300 pairwise distance matrices ($\psi = 64$, $t = 200$, single linkage), sorted by angular position and normalised to $[0, 1]$. Left: **exact hyperbolic geodesic**; Centre: **AGD** ($r = 0.551$); Right: **AGD-SS** ($r = 0.773$).

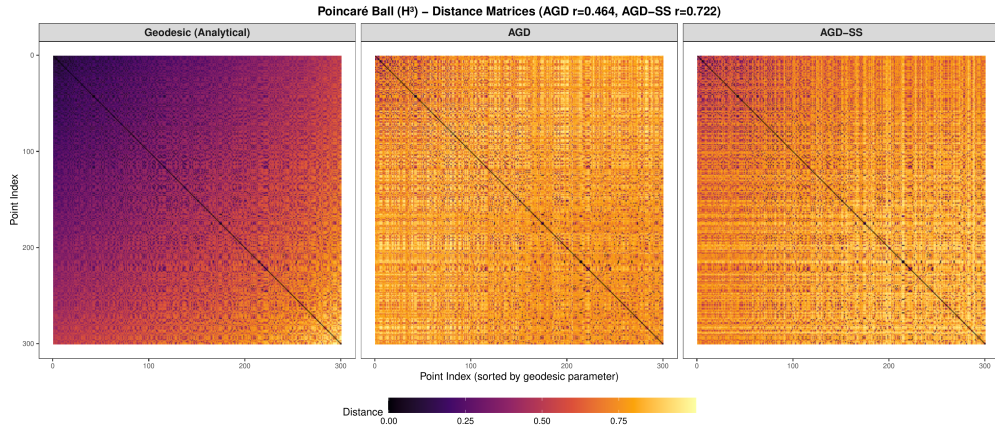


Figure 11: **Distance matrix heatmaps for Poincaré Ball (H3)**. 300×300 pairwise distance matrices ($\psi = 64$, $t = 200$, single linkage), sorted by radial distance from the origin and normalised to $[0, 1]$. Left: **exact hyperbolic geodesic**; Centre: **AGD** ($r = 0.464$); Right: **AGD-SS** ($r = 0.722$).

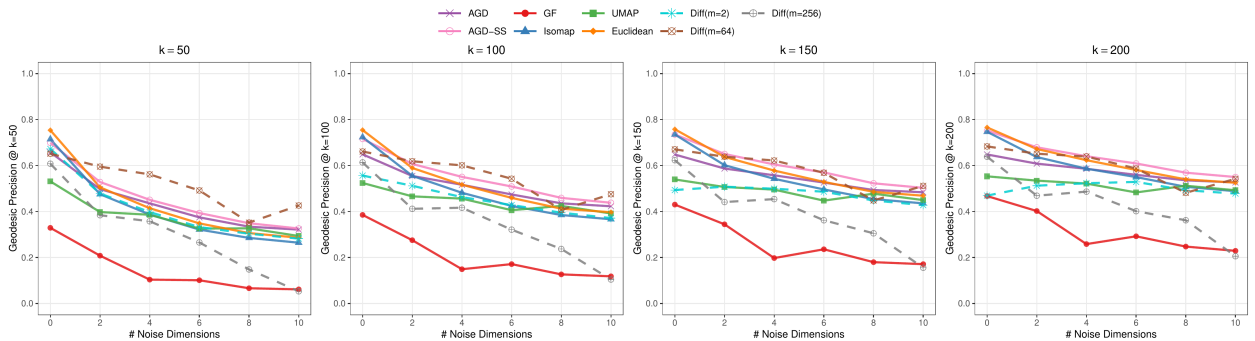


Figure 12: **Geodesic Precision on the Poincaré Ball (H3)** at $k = 50, 100, 150, 200$ ($N = 1000$, noise constant = 20). Here we use the same parameter setting as in the main paper (Figure 3). Each panel shows a different evaluation neighbourhood size k . **Diffusion** ($m = 64$) is the most noise-robust method when $k \leq 100$, but its performance collapses at higher noise levels and larger k values. **AGD-SS** performs better when k is larger ($k = 200$). **GF** performs poorly throughout (≤ 0.47), confirming its difficulty with hyperbolic geometry.

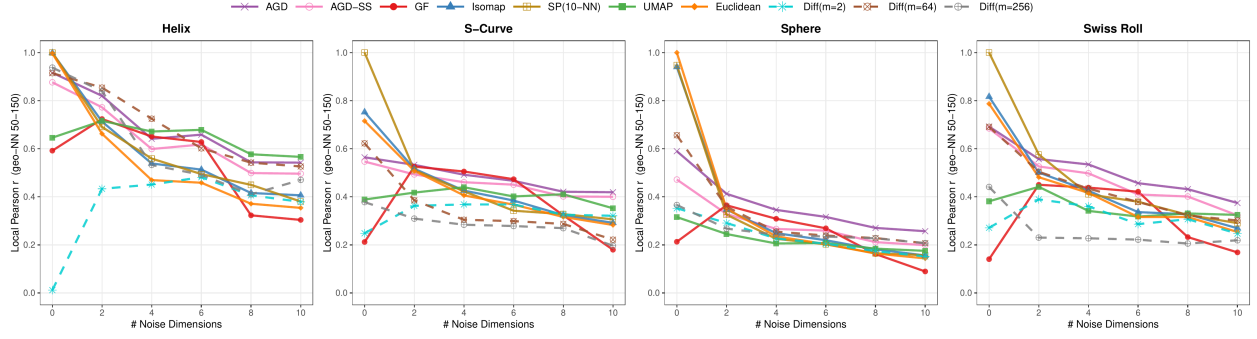


Figure 13: **Pearson correlation with true geodesic distance on four manifolds** ($N = 1000$, noise constant = 20, noise dims $\in \{0, 2, 4, 6, 8, 10\}$). In order to align with the same setting of Figure 3 in our main paper, for each point x_i we identify its 50th–150th nearest neighbours under the true geodesic (a set S_i of 101 points) and compute the Pearson r between the estimated and true geodesic distances to S_i . The plotted value is the average over all N points. Ten methods are compared: the nine from Figure 3 in the main paper, plus **SP(10-NN)**, which computes shortest-path distances on a 10-NN graph without the classical MDS embedding step used by Isomap. In Sphere and Swiss Roll, AGD-SS outperforms other methods.

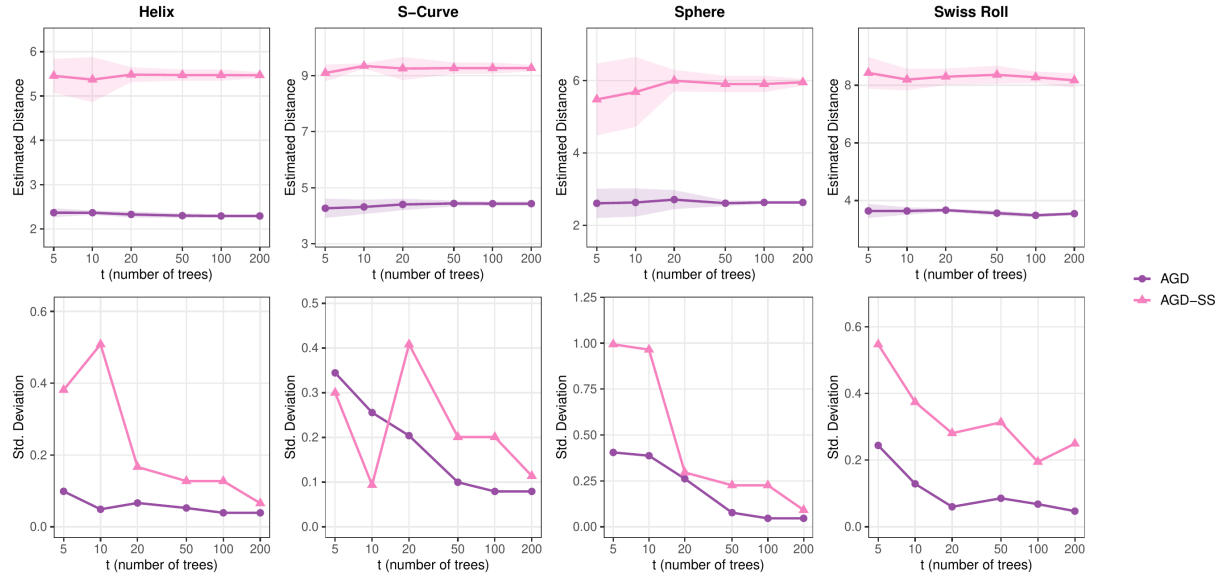


Figure 14: **Pointwise concentration of AGD and AGD-SS for a single pair of points** (5 independent runs). One pair (i, j) was randomly selected and held fixed across all runs. **Top row:** mean estimated pairwise distance $\pm$ one standard deviation (shaded). **Bottom row:** standard deviation of the estimate across 5 runs.

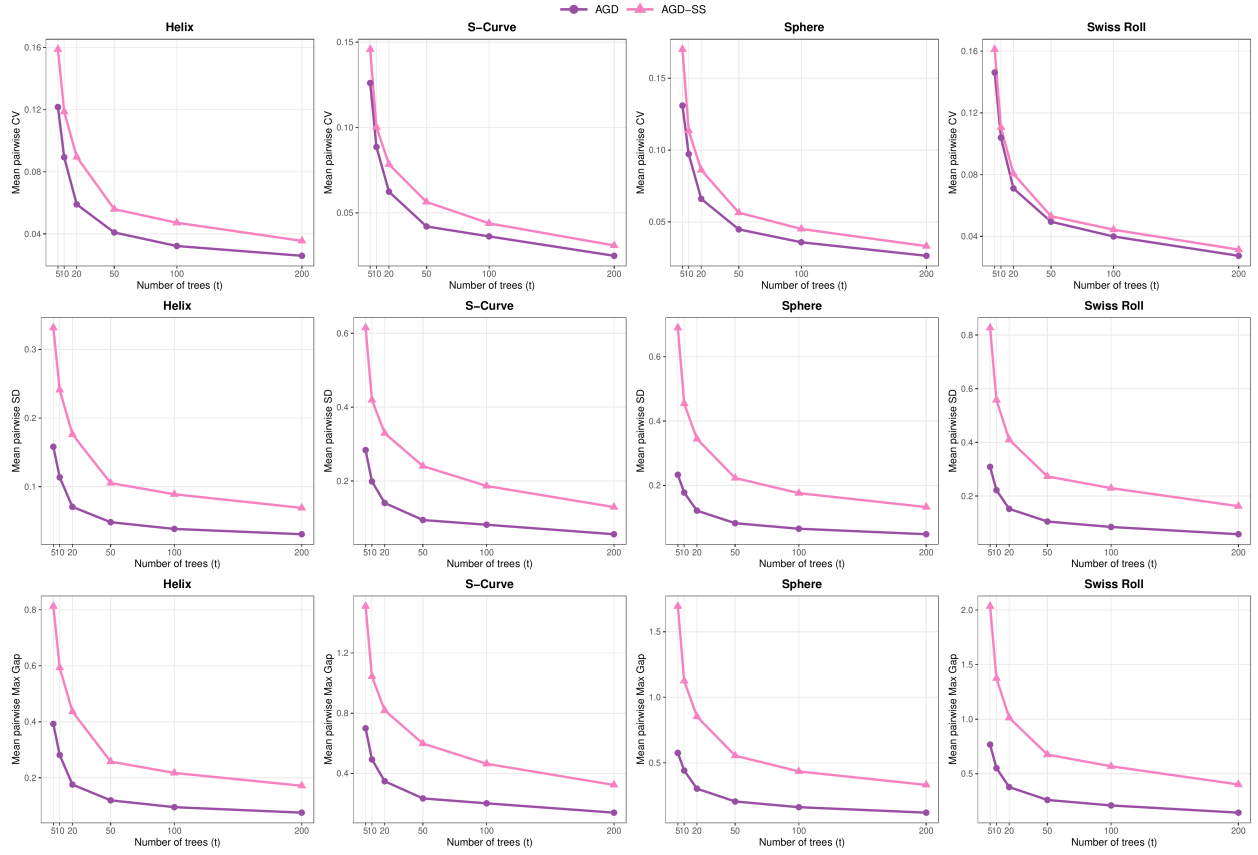


Figure 15: **Pairwise concentration across all $\binom{N}{2}$ point pairs** ($N = 300$, $\psi = 64$, single linkage, $R = 5$ independent runs). Let $\hat{d}_{ij}^{(r)}$ denote the estimated distance between points i and j in the r -th independent run ($r = 1, \dots, R$). Define the per-pair mean $\mu_{ij} = \frac{1}{R} \sum_{r=1}^R \hat{d}_{ij}^{(r)}$, standard deviation $\sigma_{ij} = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (\hat{d}_{ij}^{(r)} - \mu_{ij})^2}$, coefficient of variation $CV_{ij} = \sigma_{ij} / \mu_{ij}$, and max gap $MG_{ij} = \max_r \hat{d}_{ij}^{(r)} - \min_r \hat{d}_{ij}^{(r)}$. All statistics below are averaged over the $\binom{N}{2}$ pairs. **Top row (Mean Pairwise CV):** $\overline{CV}(t) = \left(\binom{N}{2} \right)^{-1} \sum_{i < j} CV_{ij}$, the mean coefficient of variation as a function of the number of trees t . **Middle row (Mean Pairwise SD):** $\bar{\sigma}(t) = \left(\binom{N}{2} \right)^{-1} \sum_{i < j} \sigma_{ij}$, the mean raw standard deviation (unnormalised) across all pairs. **Bottom row (Mean Pairwise Max Gap):** $\overline{MG}(t) = \left(\binom{N}{2} \right)^{-1} \sum_{i < j} MG_{ij}$, the mean of the per-pair range (maximum minus minimum) over R runs, measuring the worst-case single-run deviation for a typical pair.

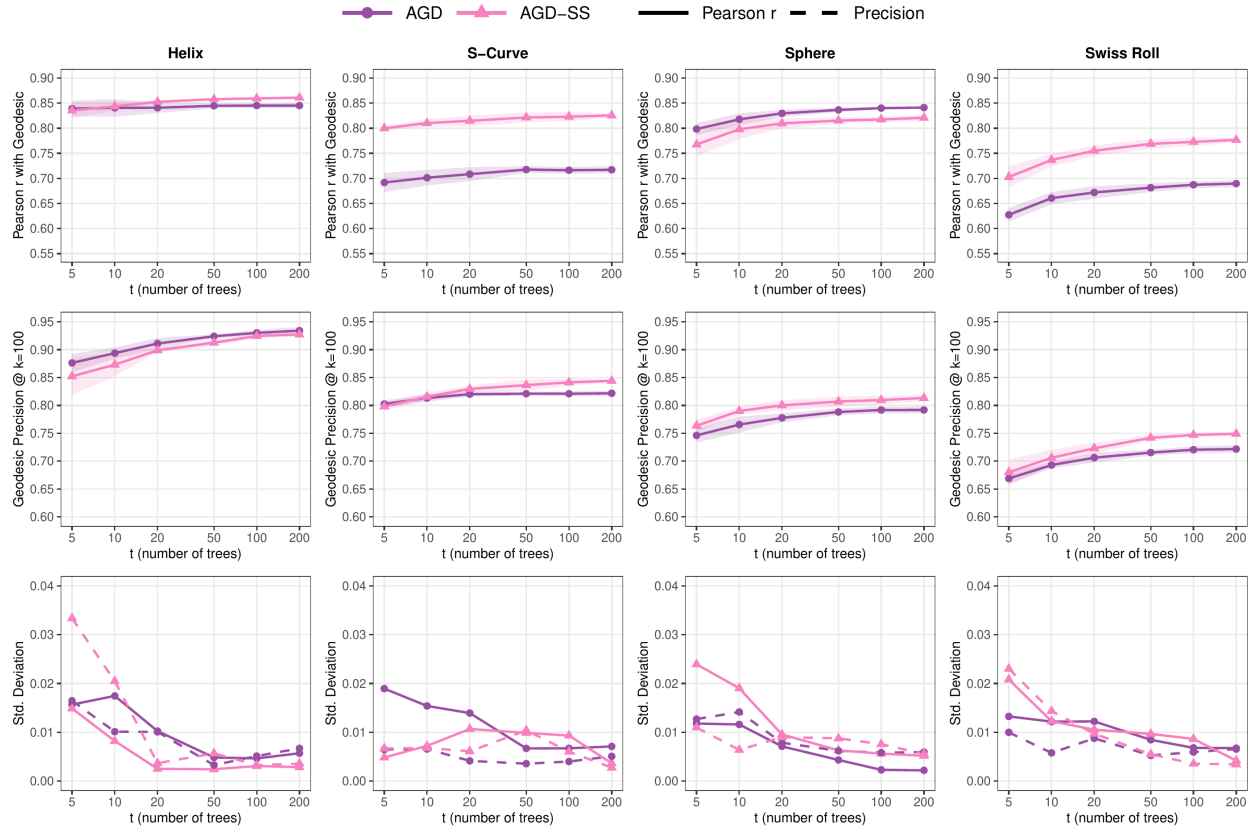


Figure 16: **Concentration of AGD and AGD-SS as a function of t (number of trees)** (5 independent runs). **Top row:** Pearson r with the true geodesic distance matrix. **Middle row:** Geodesic Precision at $k = 100$. **Bottom row:** Standard deviation across the 5 runs (solid = Pearson r , dashed = Precision). Shaded bands in the top two rows denote standard deviation.